

Testing the Weak form of Efficient Market Hypothesis: Empirical Evidence from Asia-Pacific Markets

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Abstract

This empirical study is conducted to test the weak-form market efficiency of the stock market returns of Pakistan, India, Sri Lanka, China, Korea, Hong Kong, Indonesia, Malaysia, Philippine, Singapore, Thailand, Taiwan, Japan and Australia. Monthly observations are taken for the period January 2004 to December 2009. Autocorrelation, Ljung-Box Q-statistic Test, Runs Test, Unit Root Test and the Variance Ratio are used to test the hypothesis that the stock market follows a random walk. Monthly returns are not normally distributed, because they are negatively skewed and leptokurtic. In aggregate we concluded that the monthly prices do not follow random walks in all the countries of the Asian-Pacific region. The investors can take the stream of benefits through arbitrage process from profitable opportunities across these markets.

Keywords: Weak-form Market Efficiency, Autocorrelation, Variance Ratio, random walk, Asia-Pacific

1. Introduction

Before the explanation of efficient capital market it is purposeful to match it with the perfect capital markets. Following are the necessary conditions for perfect capital markets as explained in (Copeland and Weston, 1988). Markets are frictionless means that there are no transactions cost or taxes in an economy and whole assets are completely divisible, marketable and moreover there are no constraining

regulations. Secondly there exists perfect competition in commodity and securities markets. In a given commodity market that all producers supply goods and services at lowest average cost. In the same way in securities market it interprets that all participants are price takers. Thirdly markets are considered informationally efficient which means that information attained is totally costless and it is received in the same time by whole individuals. Fourthly all individuals are considered rational but not the person who maximizes the utility.

Fama (1970) designed EMH theory with an empirical base, and distributed the Efficient Market Hypothesis into three hypotheses based on information. The efficient-market hypothesis was Ph.D. work and published in 1960 and was written by Professor Eugene Fama at University of Chicago Booth School of Business. According to EMH theory it is described that when investors face with new set of information they can overreact and some may under react to the forthcoming situation. In these scenario investors reactions are random behavior and trace a normal distribution pattern so that the net effect on market prices may not be reliably explored to make an abnormal profitable situation, when considering transaction costs i.e. commissions and spreads. This situation may perceive by an investor in a wrong manner about the market indeed, thus in actual situation the market as an aggregate is always right. If the equity market is working efficiently, the prices will show the intrinsic values of the equity and in reply, the limited savings will be allocated to the productive investment sector optimally in such a way that will provide stream of benefits to the individual investors and to the economy of the country as a whole (Copeland and Weston 1988). Rubinstein (1975) and Latham (1985) have made extension in the basic definition of market efficiency. According to them the market is efficient regarding to an information event if that information impacts no portfolio changes. Possibility is that people may not be agreeing with the conjecture of a piece of information so some can buy an asset and others may sell in such a way that the market price is not affected. If the information makes no change in prices then the market is termed as an efficient regarding to the information as Fama (1970) concluded but not by the Rubinstein (1975) or Latham (1985) sense. A number of persons have criticized this theory in various aspects.

The regulatory bodies are in continuous try to consider the best policy regarding to decrease in market interferences to the minimum level. Efficient market hypothesis and random walk theory remained popular for the last three decades. An outstanding return can be taken if there is a gap in the market information and efficiency otherwise it is impossible but only through to luck etc. The legitimacy of the random walk hypothesis has significant inference for financial theories and for strategic investment decisions therefore this subject is significant for academicians, investors and regulatory bodies. All of the above are willing to recognize the attitude of stock prices, basis of risk-return models like CAPM. For investors specially, buying and selling strategies have to be designed by considering the prices are typified by random walks or by persistence in the short run and mean deterioration in the long run. Finally it is stated that if a stock market is inefficient, the pricing apparatus may not assure the efficient allocation of capital in an economy which effects negatively to the aggregate economy.

Moreover to increase in the capabilities for involvement in the decision making process at international level and also enhance more opportunities to improve the better standards of livings of the human beings of these belongings. The focused areas of interaction and co-operation include Banking and Finance, Development of Rural Population, Science and Technology, Human and Social Development, Agriculture Sector, Energy Sector, Health and Environment.

2. Literature Review

Hypothetical understandings states that prices in an efficient market are fully representing available information communicate the broad perception of what is intended by market efficiency. Efficient Market Hypothesis is based upon the assumption that equity prices absorb speedily to the influx of latest information therefore present prices totally replicate whole existing information. On the basis of this theory, it does not seem possible to constantly perform extraordinarily in the market by applying

any sort of information that is already known by the market, and the exception is only lucky element. In EMH any news or information is defined as anything which can affect prices that are not known in the current scenario and looks randomly in future perspective. Stock market efficiency is significant concept regarding to the mechanics of the stock markets working and its performance, moreover effective participation in the development of the country's economic structure.

Fama (1965) was in the view that the statement is general and needs to be testified; moreover, it demands to build up mathematical models and formulations for market equilibrium which will be used for testing the market efficiency. Fama (1970) reported the EMH theory as a fair game model, which indicates that the investors are confident regarding to the current market price which fully replicates all available information regarding to a security. Moreover the expected returns are based upon this price which is consistent with its risk. Fama divided the empirical tests of the hypothesis into three categories based on the given information set i. weak-form EMH, ii. Semi-strong-form EMH and iii. Strong-form EMH. The Random Walk Model (RWM) is the model which assumes that subsequent price changes are sovereign and homogeneously distributed random variables and concludes that changes in future prices cannot be forecasted through historical price changes and movements. The Random Walk Model is generally used to testify the weak-form Efficient Market Hypothesis. Inefficiency indications will compel to the regulatory authorities to take compulsory steps to avoid such scenario and restructure to accurate it. As the influential effort of Fama (1970) for thirty stocks of DJIA for the period 1957 to 1962 and found no evidence; Fama and French (1988) analyzed the industry portfolio data for the period 1926 to 85 and the results of autocorrelation indicates a U type pattern against increasing returns. Lo and MacKinlay (1988) used equal and value weighted index regarding to NYSE:AMEX for the period 1962 to 1985 and strongly rejected the RWM for the entire period.

Fama (1970); Granger (1975); Hawawini (1984); Fama (1991); and Lo (1997) comprehensively tested empirically the RWM and the weak form EMH regarding to both developed and emerging economies. They all were in the support of the conclusion that there exists empirical evidence regarding to the support of EMH theory. There are number of articles that had experienced specific stock markets individually in an empirical manner moreover there are few studies that had also matched the efficiency of various stock markets. Solink (1973) examined stocks from 8 stock markets of the France, Italy, UK, Germany, Belgium, Neither land, Switzerland, Sweden and USA. The RWM shows that the deviations are lightly more apparent in European stock markets than the USA market. It is due to technical and institutional characteristics of European Capital markets. Ang and Pohlman (1978) examined fifty four stocks belonging to 5 far Eastern equity stock markets of Japan, Singapore, Australia, Hong Kong and Philippine. They found that these markets are slightly efficient in the weakest form. The reason is only due to the effect of the greater existence of extreme returns and no concern with price dependencies as explained by serial correlations. Errunza and Losq (1985) studied the behavior of equity prices of 9 emerging equity markets. The results revealed that the probability distributions are consistent with a lognormal distribution regarding to some securities showing non-stationary variance. Less developed countries (LDC) markets are less efficient than developed countries markets. The reason behind the behavior of security prices seems to be generalizable able for the severely traded segments of the less developed countries markets.

Urrutia (1995) investigated the Random Walk Model for 4 Latin American emerging stock markets. He used the monthly index data for Argentina, Brazil, Chile and Mexico for the period December 1975 to March 1991. Variance ratio test rejects the random walk hypothesis but runs test indicates that there exists weak form of efficiency regarding to these markets. The reason behind this scenario is that the domestic investors are not enough competent to design trading strategies that may allow them to earn excess returns.

Huang (1995) examined the equity markets of 9 Asian countries. He used the variance ratio statistics to test the random walk hypothesis of the Asian stock markets. He found that the RWM hypothesis for Korean and Malaysian equity market is strongly rejected for all changed holding periods. Moreover the RWM hypothesis is also rejected for the equity markets of Hong Kong,

Singapore, and Thailand. Dahel and Laabas (1999) investigated the efficiency of Bahrain, Kuwait, Saudi Arabia and Oman belonging to Gulf Cooperation Council equity markets. They investigated the observations from year 1994 to 1998. They concluded that the stock market of Kuwait is strongly in support of weak form of efficiency and other markets reject the weak form of the EMH. The reason seems to be the strong market characteristics of the Kuwait in comparison to the other three markets. Fama(1991) and Lo(1997) empirically studied and detected a number of anomalies like such as the January effect, effect of holiday, effect of weekend, the small size effect, and volatility tests. Large number of empirical tests has been applied in the literature to investigate the acceptability and validity of EMH and the RWM. Regarding to the scenario of Pakistan Hasan, Shah and Abdullah (2007) examined the weak-form market efficiency of Karachi Stock Exchange (KSE). The results reveal that prices behavior is not supporting random walks and hence these are not weak-form efficient. For such situation technical analysis may be helpful in predicting equity markets behaviors in the short run. The prior empirical findings are based upon the data of developed equity markets and hence it implies that the security prices are reacting immediately to all publicly available information.

Tests are categorized into two groups. The 1st group consists upon a contrast of risk-return results for trading or to make purify regulations that make investment decisions based on historical market information in opposition to outcome from an easy buy and hold strategy. The 2nd group engages statistical tests of independence among rates of return. Autocorrelation and Runs tests are the famous tools to test this part, Reilly and Brown (2003). The world markets initiated concentration on the study of this particular issue. There are number of studies on different individual markets as well as on regional markets e.g, Latin America Urrutia(1995). For Brazil and Mexico, Grieb and Reyes (1999), both studies are in support of random walk. Few studies about African market by Magnusson and Wydick (2000) that favors the random walk hypothesis for all the markets. Groenewold and Ariff(1999) studied ten countries in the Asia-Pacific region to evaluate the effect of liberalization on market-efficiency. They found that numerous measures of market-efficiency are unchanged by de-regulation. On the other hand methods based on regression and autocorrelation point towards greater predictability for domestically as well as internationally after de-regulation. These findings for the international circumstance may be described by the larger integration of international equity markets but the domestic phenomenon remains a mystery.

Abraham et al. (2002) studied Middle East markets. They observed that index in thinly traded equity markets may not embodied the true fundamental index value. Moreover there is a systematic bias towards rejecting the EMH. The three emerging Gulf equity markets show infrequent trading significantly that has changed the results of market efficiency and random walk tests. Worthington and Higgs (2004) investigated 20 European countries for the period August 1995 to May 2003 by applying serial correlation test, runs test for random walk, Augmented Dickey Fuller test (ADF) to test the stationarity and Lo and MacKinlay (1988) variance ratio test. They concluded that all indices are not normally distributed and only 5 countries fulfill the sternest criteria for a random walk. According to their findings Germany, Ireland, Portugal, Sweden and the United Kingdom follow random walk purely and France, Finland, Netherlands, Norway and Spain are following the random walk hypothesis. In a recent study conducted by Borges (2008) on the equity markets of France, Germany, UK, Greece, Portugal and Spain, for the period January 1993 to December 2007. They used a serial correlation test, an augmented Dickey-Fuller test, a runs test and the Lo and MacKinlay (1988) multiple variance ratio to test the random walk in equity markets. The results provide insignificant evidences that monthly prices and returns follow RWM in all six equity markets. Daily returns are abnormally distributed as indicated by the negative skeweness and leptokurtic. France, Germany, UK and Spain follow the random walk with daily data but that hypothesis rejects random walk hypothesis for Greece and Portugal. The reason is due to serial positive correlation. But after year 2003 these two countries also follows random walk behavior.

No doubt that there are number of studies on the efficient market hypothesis to test the randomness of stock prices of individual companies but still there are enough gaps in the study regarding to test the random walk of equity market indices around the globe in present era. Therefore the Asian-Pacific markets have been selected to test the market efficiency of various emerging and developed markets in the region.

3. Data and Methodology

The observations are monthly closing values of stock market indices for 14 Asia-Pacific countries including Pakistan, India, Sri Lanka, Indonesia, Malaysia, Thailand, Taiwan, Hong Kong, Singapore, Philippine, China, Korea, Japan, and Australia. Observations are taken for the period January 2004 to 31 December 2009. Market returns are computed as follows.

$$R_t = \ln(P_t / P_{t-1}) \quad (1)$$

P_t = Market Price at time 't'

P_{t-1} = Market Price at time 't-1'

3.1. Descriptive Statistics

Descriptive Statistics for the stock returns includes the Arithmetic Mean, Median, Standard Deviation Jarque-Bera, Variance, Kurtosis, Skewness, and Range. The Jarque-Bera statistics is used to test the normality of the data series.

3.2. Auto Correlation and Ljung Box Statistics

The serial autocorrelation is used to test the relationship between the time series its own values at different lags. If the serial autocorrelation is negative it means it is mean reverting and accepts the null hypothesis and if the result is positive coefficients then it rejects the null hypothesis. Another technique that will be use is Ljung-Box. Ljung-Box test provides a superior fit to the chi-square (χ^2) distribution for little samples.

$$Q_{Ljung-Box} = n(n+2) \sum_{t=1}^k \frac{\psi^2(t)}{n-t} \quad (2)$$

3.3. Runs Test

We apply runs test to analyze the serial independence in the returns stream which search out whether succeeding price variations are autonomous to each other as it happens under the random walk null hypothesis. If the number of runs are being observed and the forthcoming price variations (or returns change) with the similar sign. In a series of consecutive price variations (or returns change) the null hypothesis can be tested. We can take into consideration two approaches i.e., positive return (+) which means that return > 0 and secondly a negative return (-) which means that returns < 0 and it is based on with respect to mean return. Second consideration has the benefit of permitting for and to accurate the impact and effect of an ultimate time drift in the return series. It is notable element that it is a non-parametric test and does not entail the normally distributed returns. The runs test stands upon the argument that if price changes or returns are random then actual number of runs (*Runs*) must be near to the expected number of runs. Let $+m$ and $-m$ are reflecting the totality of positive returns (+) and totality of negative returns (-) regarding to a sample with " m " observations, where $m = m_+ + m_-$. For greater sample size the test statistic is just about normally distributed:

$$Z = \frac{\omega - \mu_\omega}{\sigma_\omega} \approx N(0,1) \quad (3)$$

Where

$$\mu_\omega = \frac{2m_+m_-}{m} + 1 \text{ and } \sigma_\omega = \sqrt{\frac{2m_+m_-(2m_+m_- - m)}{m^2(m-1)}}$$

3.4. Unit Root Tests

Augmented Dickey-Fuller (ADF) test is applied to test the presence of unit root in the time series of stock price changes in the indices. Majorly it is used to test the stationarity of the time series. It is inferred from the OLS as follows:

$$\Delta R_t = b_0 + b_1 + \pi_0 R_{t-1} + \sum_{i=1}^j \psi_i \Delta R_{it-1} + \epsilon_t \quad (4)$$

R_t is the price at time t ,

ΔR_t = change in price

3.5. Variance Ratio Tests

A significant assumption of the random walk theory is investigated through variance ratio test. If R_t is a random walk then the ratio of the variance of the j^{th} difference scaled by “j” to the σ^2 of the first difference have a propensity equal to one, that is why the σ^2 of the j-differences boosts linearly in the surveillance interval,

$$VR(j) = \frac{\sigma^2(j)}{\sigma^2(1)} \quad (5)$$

$\sigma^2(j) = 1/j^{\text{th}}$ variance of the j-differences.

$\sigma(1)$ = is the variance of the first differences. For H_0 the $VR(j)$ must move toward unity.

MacKinlay [1988] used the following formula and proposed the specification test for a given sample size of $mj+1$ observations.

$$\sigma^2(j) = \frac{1}{k} \sum_{t=j}^{mj} (R_t - R_{t-1} - j\hat{\mu})^2 \quad (6)$$

Where $k = j(mj - j + 1)[1 - \frac{j}{mj}]$ and $\hat{\mu}$ shows the mean of the sample.

$$(R_t - R_{t-1}): \hat{\mu} = \frac{1}{mj} (R_{mj} - R_0) \text{ and } \sigma^2(1) = \frac{1}{(mj-1)} \sum_{t=1}^{mj} (R_t - R_{t-1} - \hat{\mu})^2 \quad (7)$$

Lo and MacKinlay (1988) created the asymptotic distribution of the predicted variance ratios and recommended two test statistics, $Z(j)$ and $Z^*(j)$, under the H_0 of homo-skedastic increase random walk and hetero-skedastic increase random walk correspondingly. If the H_0 is proves true, the connected test statistics has standardized asymptotic normal distribution. By assuming homo-skedastic increments shocks, therefore we have

$$Z(j) = \frac{VR(j)-1}{\pi_o(j)} \approx N(0,1) \quad (8)$$

Where $\pi_o(j) = \left[\frac{2(2j-1)(j-1)}{3j(mj)} \right]^{1/2}$ with an assumption that there is hetroskedastic growth, the test statistics is

$$Z^*(j) = \frac{VR(j)-1}{\pi_i(j)} \approx N(0,1) \quad (9)$$

Where

$$\pi_o(j) = \left[4 \sum_{t=1}^{j-1} \left(1 - \frac{t}{j}\right) \hat{\sigma}_t \right]^{1/2} \text{ and } \hat{\sigma}_t = \frac{\sum_{i=t+1}^{mj} (R_t - R_{t-1} - j\hat{\mu})^2 (R_{i-t} - R_{i-t-1} - j\hat{\mu})^2}{\left[\sum_{i=1}^{mj} (R_t - R_{t-1} - j\hat{\mu})^2 \right]^2}$$

that is vigorous under hetroskedastic assumption, therefore it can be used for a larger time series analysis. The modus operandi suggested by Lo and MacKinlay (1988) is developed to test single variance ratio tests for a explicitly j-difference. While for the random walk hypothesis there must have $VR(j)$ equal to 1 for all j. Chow and Denning (1993) suggested multiple variance ratio test (MVRT). Take a single set of n variance ratio tests linked with the set in aggregate by interval. There are multiple sub-hypotheses under the random walk hypothesis. The rejection of single or more therefore rejects null hypothesis of the random walk. Simply to assist contrast of this study with preceding research (Lo and MacKinlay (1988) and Campbell, Lo and Mackinlay, (1997) on other equity markets, the j is chosen as 2, 4, 8, 10 and 16.

For a given set of test statistics $\{Z(j) \mid k = 1, 2, \dots, n\}$, the random walk hypothesis is not accepted if any one of the $VR(j_k)$ is considerably dissimilar than one and the only maximum absolute value in the given set of test statistics is taken. Chow and Denning (1993) multiple variance ratio test (MVRT) is based on this result:

$$MN\{\max(|Z(j_1)|, \dots, |Z(j_k)|) \leq SM(\gamma; n; T)\} \geq 1 - \gamma \quad (10)$$

$SM(\gamma; n; T)$ = is the higher γ position of the Studentize Maximum Modulus (SM) distribution with constraints “ n ” and T sample size degrees of freedom.

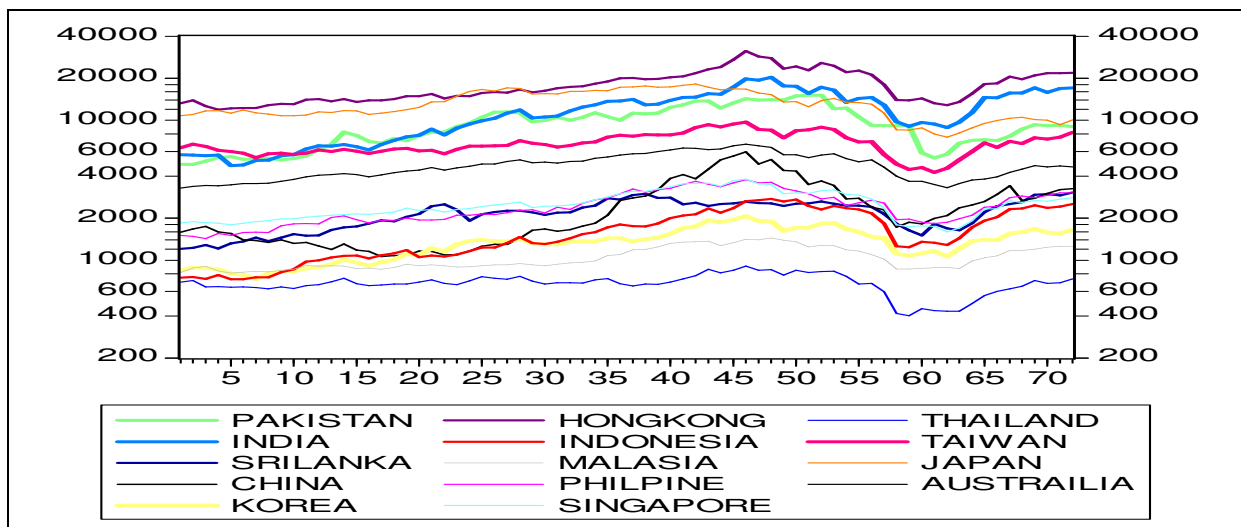
$$\text{Asymptotically, } \lim_{T \rightarrow \infty} SM(\gamma; n; \infty) = Z_{\gamma/2} \quad (11)$$

Where $\gamma^* 2Z$ is standard normal with $\gamma^* = 1 - (1 - \gamma)1/m$. The size of the multiple variance ratio test is controlled by Chow *et al.* (1993) by comparing the computed values of the standardized test statistics, either $Z(j)$ or $Z^*(j)$ with the SM critical values. If the maximum absolute value of $Z(j) >$ the critical value at a pre arranged worth level then the random walk hypothesis is not accepted.

4. Results and Discussion

The data comprises of monthly closing values of stock market indexes for Pakistan, India, Sri Lanka, China, Korea, Hong Kong, Indonesia, Malaysia, Philippine, Singapore, Thailand, Taiwan, Japan and Australia. The data includes monthly observations from January 2004 to December 2009, during which some of these markets remained volatile, especially in the case of Pakistan, India, China, Sri Lanka, Hong Kong, Indonesia, Korea, Thailand and Taiwan as shown in Figure 1.

Figure 1: Trend of Asia-Pacific Equity Market Indices



Trend shows that the prices are moving cumulatively in a systematic manner. Table 1 shows the descriptive statistics for the returns of the Asian-Pacific equity market indices. The monthly returns are negatively skewed in all 14 countries for the period 2004 to 2009 which indicates that large negative returns (minimum extreme value) are larger than the higher positive returns (maximum extreme value). The kurtosis is positive for all countries which mean that the distributions of returns are leptokurtic indicating higher peaks than expected from normal distributions. The Indonesian market is providing the highest 1.7% return with 7.9% standard deviation. The Indian market is providing 1.5% with a standard deviation of 8.5%. The market of Sri Lanka is providing 1.3% return with a risk level of 7.1% and the equity market of Pakistan is providing 0.9% return with a risk level of 9.2%. Jarque-Bera test rejects the hypothesis of a normal distribution for Pakistan, India, Korea, Hong Kong, Indonesia, Malaysia, Philippine, Singapore, Thailand, Japan and Australia.

Table 1: Descriptive Statistics for the Asian-Pacific Market Returns

	PAK	IND	SRI	CHI	KOR	HK	INDO	MAL	PHIL	SING	THA	TAI	JAP	AUS
Mean	0.009	0.015	0.013	0.01	0.009	0.007	0.017	0.006	0.01	0.006	0.001	0.004	-0.001	0.005
Median	0.01	0.02	0.02	0.03	0.01	0.02	0.03	0.01	0.02	0.02	0.01	0.01	0.00	0.02
Maximum	0.20	0.25	0.19	0.24	0.13	0.16	0.18	0.13	0.14	0.19	0.13	0.14	0.10	0.07
Minimum	-0.45	-0.27	-0.18	-0.28	-0.26	-0.25	-0.38	-0.17	-0.28	-0.27	-0.36	-0.21	-0.27	-0.15
Std. Dev.	0.09	0.08	0.07	0.10	0.07	0.07	0.08	0.04	0.06	0.06	0.07	0.07	0.06	0.04
Skewness	-1.90	-0.76	-0.15	-0.76	-0.93	-0.89	-1.87	-0.64	-1.23	-1.26	-1.79	-0.50	-1.45	-1.42
Kurtosis	10.61	4.63	3.55	3.71	5.38	5.34	10.54	5.45	7.28	8.64	10.40	3.63	7.45	5.20
Jarque-Bera	214.10	14.65	1.14	8.24	27.12	25.42	209.67	22.50	72.26	112.93	199.80	4.10	83.57	38.25
Probabilit	0.00	0.00	0.56	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.13	0.00	0.00
y Sum	0.62	1.10	0.93	0.71	0.67	0.50	1.21	0.43	0.70	0.42	0.05	0.25	-0.06	0.35
S.Sq. Dev.	0.59	0.51	0.35	0.74	0.32	0.34	0.43	0.14	0.29	0.28	0.36	0.33	0.26	0.13
Observations	71.00	71.00	71.00	71.00	71.00	71.00	71.00	71.00	71.00	71.00	71.00	71.00	71.00	71.00

Significant at 5% level

The test accepts the hypothesis of normal distribution only for Sri Lanka, China and Taiwan for the period 2004-2009. The zero p-values of returns with respect to Jarque-Bera statistics shows that the series of returns do not follow the normal distributions. To further analyze the randomness of the return series we used serial autocorrelation and Ljung-Box Q-statistics.

If P-value < 0.05 of the Q-Statistics, and the null of the entire autocorrelation coefficients together equal to zero may be rejected at 0.05 level of significance. Therefore it is inferred that the historical returns can be used to predict future returns and this element indicates that the weak form of market efficiency does not hold. The P-values in Table 2 at first difference indicates that the null is rejected for all markets. From lag 6 to onward the equity market of Pakistan shows little efficiency. Basically the null hypothesis for random walk is rejected if the serial correlation contains the positive coefficients over different lags. If we visualize the autocorrelations at lag 1 which are negative for all the markets but hence over different lags it have positive values so we cannot infer that a market is a weak form efficient. The further analysis requires that whether the time series is non-stationary or stationary.

Table 2: Autocorrelation and Q-Statistics for Returns

Pakistan		1	2	3	4	5	6	7	8	9	10
India	AC	-0.346	-0.128	-0.124	0.075	0.061	-0.014	0.072	-0.162	0.13	-0.119
	Q-Stat	8.7509	9.9743	11.139	11.57	11.856	11.871	12.289	14.417	15.804	16.986
	Prob	0.003	0.007	0.011	0.021	0.037	0.065	0.091	0.072	0.071	0.075
Sri Lanka	AC	-0.447	-0.033	-0.101	0.122	0.104	-0.267	0.168	-0.109	0.082	0.003
	Q-Stat	14.591	14.674	15.438	16.581	17.412	23.045	25.299	26.268	26.82	26.821
	Prob	0	0.001	0.001	0.002	0.004	0.001	0.001	0.001	0.001	0.003
China	AC	-0.444	0.031	-0.215	0.194	-0.061	0.082	-0.074	-0.019	0.002	0.037
	Q-Stat	14.4	14.472	17.936	20.8	21.091	21.627	22.06	22.088	22.089	22.205
	Prob	0	0.001	0	0	0.001	0.001	0.002	0.005	0.009	0.014
Korea	AC	-0.614	0.235	-0.25	0.247	-0.006	-0.211	0.254	-0.224	0.254	-0.238
	Q-Stat	27.546	31.641	36.359	41.023	41.025	44.533	49.677	53.745	59.091	63.83
	Prob	0	0	0	0	0	0	0	0	0	0
Hong Kong	AC	-0.548	0.128	-0.122	0.187	-0.01	-0.249	0.246	-0.199	0.13	-0.03
	Q-Stat	21.949	23.17	24.287	26.954	26.963	31.85	36.7	39.907	41.307	41.382
	Prob	0	0	0	0	0	0	0	0	0	0
Indonesia	AC	-0.461	0.071	-0.162	0.101	0.087	-0.208	0.174	-0.19	0.158	0.041
	Q-Stat	15.511	15.881	17.859	18.645	19.233	22.641	25.06	27.999	30.051	30.189
	Prob	0	0	0	0.001	0.002	0.001	0.001	0	0	0.001
Malaysia	AC	-0.355	-0.169	0.036	0.15	-0.094	-0.138	0.228	-0.116	-0.069	0.107
	Q-Stat	9.2235	11.331	11.429	13.137	13.827	15.338	19.491	20.584	20.982	21.935
	Prob	0.002	0.003	0.01	0.011	0.017	0.018	0.007	0.008	0.013	0.015
Philippine	AC	-0.463	-0.013	0.084	-0.133	0.147	-0.237	0.191	-0.083	0.004	-0.011
	Q-Stat	15.69	15.702	16.231	17.574	19.258	23.685	26.607	27.165	27.166	27.176
	Prob	0	0	0.001	0.001	0.002	0.001	0	0.001	0.001	0.002
Singapore	AC	-0.54	0.06	-0.071	0.14	-0.106	0.17	-0.196	0.061	0.067	-0.065
	Q-Stat	21.274	21.543	21.928	23.418	24.294	26.557	29.632	29.937	30.304	30.664
	Prob	0	0	0	0	0	0	0	0	0	0.001
Singapore	AC	-0.453	0.129	-0.268	0.183	0.089	-0.173	0.04	-0.093	0.169	-0.109
	Q-Stat	14.981	16.214	21.603	24.156	24.777	27.124	27.253	27.95	30.319	31.321

	Prob	0	0	0	0	0	0	0	0	0	0.001
Thailand	AC	-0.321	-0.331	0.146	0.123	-0.031	-0.184	0.174	-0.054	-0.056	0.078
	Q-Stat	7.5109	15.613	17.222	18.377	18.45	21.104	23.531	23.765	24.02	24.527
Taiwan	Prob	0.006	0	0.001	0.001	0.002	0.002	0.001	0.003	0.004	0.006
	AC	-0.512	0.132	-0.093	0.033	0.037	-0.085	0.011	-0.023	-0.068	0.21
	Q-Stat	19.161	20.449	21.099	21.18	21.284	21.859	21.869	21.912	22.293	25.989
	Prob	0	0	0	0	0.001	0.001	0.003	0.005	0.008	0.004
Japan	AC	-0.313	-0.223	0.06	0.238	-0.192	-0.131	0.147	-0.062	0.022	-0.016
	Q-Stat	7.1733	10.853	11.122	15.432	18.283	19.641	21.373	21.689	21.728	21.751
	Prob	0.007	0.004	0.011	0.004	0.003	0.003	0.003	0.006	0.01	0.016
	AC	-0.327	-0.19	0.016	0.24	-0.208	-0.087	0.118	0.023	-0.049	-0.009
Australia	Q-Stat	7.8198	10.485	10.504	14.92	18.276	18.87	19.977	20.022	20.223	20.23
	Prob	0.005	0.005	0.015	0.005	0.003	0.004	0.006	0.01	0.017	0.027

Significant at 5% level

So the unit root test is applied to check the stationarity as a necessary condition for Random walk. According to the Random walk hypothesis the log price series must have a unit root whereas the returns series must be stationary. For this purpose the Augmented Dickey-Fuller Test (1981) is used to test the stationary of the time series.

Table 3: Unit Root Test

Equity Markets	Augmented Dicky-Fuller Test at Level	Augmented Dicky-Fuller Test at 1st Difference
PAKISTAN	-1.68496	-7.1851***
INDIA	-1.0552	-7.6373***
SRI LANKA	-1.35201	-6.7887***
CHINA	-1.50308	-4.0473***
KOREA	-1.38661	-7.8845***
HONG KONG	-1.33181	-6.9908***
INDONESIA	-1.19877	-5.9370***
MALAYSIA	-1.2097	-6.5838***
PHILPINE	-1.32638	-7.6034***
SINGAPORE	-2.05654	-6.3217***
THAILAND	-1.72684	-7.1242***
TAIWAN	-1.39732	-7.1774***
JAPAN	-1.22371	-6.3806***
AUSTRALIA	-1.6934	-5.7152***

According to Table 3, the time series of indices is non-stationary at order $I(0)$ and it becomes stationary for order $I(1)$ at 1% and 5 % level of significance. After unit root test we further applied the runs test. The results of the runs test do not depend upon the normality of returns are displayed in Table 4. Runs test is defined as the series of consecutive price changes with the identical sign. The H_0 elucidates that the succeeding price changes are not dependent and moves randomly.

Table 4: Runs Test at $K = \text{Mean Return}$

	PAK	IND	SRI	CHI	KOR	HK	INDO	MAL	PHIL	SING	THA	TAI	JAP	AUS
K= Mean	.0087	.016	.013	.0100	.0094	.007	.017	.0060	.0098	.0058	.0007	.0035	-.001	.005
Cases<K	35	33	34	29	33	28	29	31	32	27	32	33	32	26
Cases ≥K	36	38	37	42	38	43	42	40	39	44	39	38	39	45
Total	71	71	71	71	71	71	71	71	71	71	71	71	71	71
Cases														
No of														
Runs	31	39	34	31	33	38	34	36	39	33	41	35	31	26
Z	-1.31	.643	-.584	-1.06	-.799	.772	-.324	.017	.687	-.372	1.170	-.318	-1.24	-2.05
P-value	.189	.520	.560	.286	.424	.440	.746	.986	.492	.710	.242	.750	.213	.040*

Significant at 5% level

Table 4.1: Runs Test at K = 0

	PAK	IND	SRI	CHI	KOR	HK	INDO	MAL	PHIL	SING	THA	TAI	JAP	AUS
K= 0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Cases<K	27	24	26	27	31	25	25	26	28	25	31	32	33	24
Cases ≥K	44	47	45	44	40	46	46	45	43	46	40	39	38	47
Total Cases	71	71	71	71	71	71	71	71	71	71	71	71	71	71
No of Runs	35	32	31	29	33	35	31	31	39	29	39	37	33	24
Z	.136	-.20	-.763	-1.38	-.712	.421	-.628	-.763	1.023	-1.15	.746	.204	-.799	-2.34
P-value	.892	.836	.446	.165	.476	.674	.530	.446	.306	.249	.456	.838	.424	.019*

Significant at 5% level

During the period 2004-2009, the total cases of runs is significantly less than the expected number of runs for all the countries and the Australia at K = Mean Value as well as K = 0 have least expected number of runs against total cases so all markets clearly rejects the random walk hypothesis. However, these results must be testified by using the most modern Variance Ratio test introduced by Lo and MacKinlay (1988). If the Variance Ratio test statistic > 1 , then the series is positively correlated. In our study it does not holds true for all countries. In the case of Pakistan at $j=2$, the (Variance Ratio – 1) returns the value of Auto Correlation Function at lag 1, it can be hence proved with the Auto Correlation Function as given in the Table 2 and the variance ratio given in Table 5 respectively. Lets we take the VR (j) 0.67 for Pakistan at lag 2. By subtracting 0.67 to 1 we get the value of -0.33 which is reflecting in table 2 for Pakistan at lag 1. In the same way by taking VR (j) 0.56 for India and subtracting 1 we get the value -0.44 which reflects the authenticity of the results.

Table 5: Variance Ratio Test at Return Series

Country	Period = J	2	4	8	16
Pakistan	VR(J)	0.671214	0.312232	0.212335	0.088846
	z(j)	-3.91794	-6.19537	-6.34617	-6.97686
	z*(j)	-1.60224	-2.06281	-1.73837	-1.49133
	Probability	0.1091	0.0391	0.0821	0.1359
INDIA	VR(J)	0.55922	0.248223	0.164836	0.093889
	z(j)	-5.2525	-6.77196	-6.72887	-6.93825
	z*(j)	-3.28462	-3.06844	-2.18985	-1.63763
	Probability	0.001	0.0022	0.0285	0.1015
Sri Lanka	VR(J)	0.562848	0.256469	0.177101	0.077399
	z(j)	-5.20927	-6.69768	-6.63005	-7.06451
	z*(j)	-2.72292	-2.69661	-2.09406	-1.77497
	Probability	0.0065	0.007	0.0363	0.0759
China	VR(J)	0.39669	0.193133	0.162325	0.103917
	z(j)	-7.18927	-7.2682	-6.7491	-6.86146
	z*(j)	-3.67323	-2.85177	-1.99538	-1.48258
	Probability	0.0002	0.0043	0.046	0.1382
Korea	VR(J)	0.463645	0.233409	0.168727	0.09887
	z(j)	-6.39141	-6.9054	-6.69752	-6.90011
	z*(j)	-3.56539	-2.97348	-2.14914	-1.62098
	Probability	0.0004	0.0029	0.0316	0.105
Hong Kong	VR(J)	0.536949	0.304217	0.205704	0.125296
	z(j)	-5.51789	-6.26757	-6.3996	-6.69776
	z*(j)	-3.44112	-2.5955	-1.85527	-1.40908
	Probability	0.0006	0.0094	0.0636	0.1588
Indonesia	VR(J)	0.663698	0.329224	0.228903	0.119158
	z(j)	-4.0075	-6.04231	-6.21268	-6.74476
	z*(j)	-1.99193	-2.35567	-1.9285	-1.58618
	Probability	0.0464	0.0185	0.0538	0.1127
Malaysia	VR(J)	0.548211	0.324723	0.204798	0.123367
	z(j)	-5.38369	-6.08285	-6.40689	-6.71253
	z*(j)	-3.40056	-2.98971	-2.28213	-1.65
	Probability	0.0007	0.0028	0.0225	0.0989
Philippine	VR(J)	0.470093	0.22762	0.134749	0.101634

Singapore	$z(j)$	-6.31457	-6.95755	-6.97128	-6.87894
	$z^*(j)$	-3.09248	-2.73009	-2.11396	-1.58174
	Probability	0.002	0.0063	0.0345	0.1137
	VR(J)	0.560729	0.330287	0.216199	0.130152
	$z(j)$	-5.23452	-6.03273	-6.31504	-6.66057
Thailand	$z^*(j)$	-2.66363	-2.301	-1.77156	-1.36371
	Probability	0.0077	0.0214	0.0765	0.1727
	VR(J)	0.686842	0.272052	0.205264	0.107518
	$z(j)$	-3.73171	-6.55731	-6.40314	-6.83389
	$z^*(j)$	-1.64928	-2.17609	-1.69175	-1.42171
Taiwan	Probability	0.0991	0.0295	0.0907	0.1551
	VR(J)	0.496617	0.313721	0.193852	0.114967
	$z(j)$	-5.9985	-6.18196	-6.49509	-6.77685
	$z^*(j)$	-3.50936	-2.64898	-2.02365	-1.54316
	Probability	0.0004	0.0081	0.043	0.1228
Japan	VR(J)	0.649095	0.302877	0.202883	0.117857
	$z(j)$	-4.18152	-6.27964	-6.42232	-6.75472
	$z^*(j)$	-2.47675	-2.48952	-1.88323	-1.47331
	Probability	0.0133	0.0128	0.0597	0.1407
	VR(J)	0.68341	0.339093	0.232623	0.126043
Australia	$z(j)$	-3.77261	-5.95341	-6.18271	-6.69204
	$z^*(j)$	-2.2719	-2.45304	-1.75768	-1.35968
	Probability	0.0231	0.0142	0.0788	0.1739

The standardized VR test statistics for $z(j)$ and $z^*(j)$ is significant at $J = 2$, $J=4$ and $J=8$ for all countries except Japan, Australia and Pakistan. For Japan and Australia it is significant for $j=2$ and $J=4$ but for Pakistan it is significant only for $j=4$. An important observation in the above cases is that, as the variance ratio increases with j , the $z(j)$ and $z^*(j)$ also increase in most cases which indicates that as ' j ' increases, the significance of the rejection becomes stronger.

According to the variance ratio test it is inferred that the equity market of the Asian-Pacific region remained inefficient for the period 2004-2009. After whole discussion it is worth noting that the acceptance or rejection of the Random Walk Hypothesis does not essentially entails that the equity markets are efficient or inefficient respectively (Lo and MacKinlay, 1988), because the conclusions of this research are based on samples.

SUMMARY TABLE

Do Asian Pacific Equity Markets Follows Random Walk

COUNTRY	Serial Autocorrelation	Ljung- Box Q-static	Unit Root Test At First Difference	Runs Test AT $k=\text{mean}$ and $k=0$	Variance Ratio at Return
PAKISTAN	NO	NO	YES	NO	NO
INDIA	NO	NO	YES	NO	NO
SRI LANKA	NO	NO	YES	NO	NO
CHINA	NO	NO	YES	NO	NO
KOREA	NO	NO	YES	NO	NO
HONG KONG	NO	NO	YES	NO	NO
INDONESIA	NO	NO	YES	NO	NO
MALAYSIA	NO	NO	YES	NO	NO
PHILIPPINE	NO	NO	YES	NO	NO
SINGAPORE	NO	NO	YES	NO	NO
THAILAND	NO	NO	YES	NO	NO
TAIWAN	NO	NO	YES	NO	NO
JAPAN	NO	NO	YES	NO	NO
AUSTRALIA	NO	NO	YES	NO	NO

The summary table indicates that no one market completely follows the random walk hypothesis and hence these markets remained inefficient throughout this time period.

5. Conclusion

This empirical study investigates the weak form of market efficiency in the Asia-Pacific region. The sample size consists of 14 equity markets from this region. The purpose of the study is to investigate whether the selected equity markets follows the Random Walk Model at individual level or not. No arbitrage profits can be earned if the equity markets are efficient at individual level. To verify the normal distribution of the data we performed Jarque-Bera test and visualized the skewness and kurtosis. The results reveal that the Jarque-Bera test rejects the hypothesis of a normal distribution for Pakistan, India, Korea, Hong Kong, Indonesia, Malaysia, Philippine, Singapore, Thailand, Japan and Australia. The skewness indicates that the data is negatively skewed for all the countries. To verify the weak form of efficiency, Runs Test, Unit Root Test, Autocorrelation, Ljung-Box Q-Statistic and Variance Ratio tests were applied for this purpose. By applying unit root test the results reveal that the data series become stationary at order I (1). Finally the results of the Autocorrelation, Ljung-Box Q-Statistic and Variance Ratio tests indicates that no one market is weak form efficient and strongly rejects the null hypothesis. Hence it is concluded that the investors may get the stream of arbitrage benefits due to market inefficiency belonging to these countries.

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